

Lemma NT in-bundle: the circle representation bound $r_{L_m}(D) \leq C d(D)$ for rank-2 lattice sections

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

Throughout, $m \in \mathbb{Z}^3 \setminus \{0\}$, $m_0 = m / \gcd(m)$ its primitive part, and $L_m = \mathbb{Z}^3 \cap m^\perp = L_{m_0}$ the rank-two lattice section. For an integer $D \geq 1$ write $r_{L_m}(D) = \#\{y \in L_m : |y|^2 = D\}$ and let $d(\cdot)$ be the divisor function. The DR-2 lattice theorem uses $r_Q(m) \leq \#(Q \cap C_m) \leq r_{L_m}(4R - |m|^2)$ and needs $r_{L_m}(D) \ll_\varepsilon D^\varepsilon$, uniformly in m .

Lemma NT (two-tier pin). *(i) (Unramified case; PROVED in full below.) Suppose every odd prime $p \mid D$ satisfies $p^2 \nmid \Delta_0$, where Δ_0 is the discriminant of the primitive part of the Gram form of L_{m_0} . Then*

$$r_{L_m}(D) \leq 24 d(D)^2 \ll_\varepsilon D^\varepsilon.$$

(ii) (Ramified corner; citation + outline + exhaustive applied-domain machine coverage.) Without the unramified hypothesis the same conclusion $r_{L_m}(D) \ll_\varepsilon D^\varepsilon$ holds, with the classical sharp form $r_{L_m}(D) \leq 6 d(D)$ (Dirichlet); see Sec. 5.

Scope. Tier (i) covers the generic situation and is self-contained. Tier (ii) is the only imported step; on the APPLIED domain of the DR-2 package (all six shells $R = 101\text{--}9974$) it is verified exhaustively by machine for every single sum m (Sec. 5), so no applied instance rests on the citation alone.

2. Setup: the section form, content, and 2-adic control

Fix a basis (u, v) of L_{m_0} . The length form $q(x, y) = |xu + yv|^2 = \alpha x^2 + \beta xy + \gamma y^2$ has integer coefficients ($\alpha = |u|^2$, $\beta = 2u \cdot v$, $\gamma = |v|^2$) and discriminant

$$\Delta = \beta^2 - 4\alpha\gamma = -4 \det \begin{pmatrix} |u|^2 & u \cdot v \\ u \cdot v & |v|^2 \end{pmatrix} = -4 \operatorname{covol}(L_{m_0})^2 = -4|m_0|^2,$$

since a primitive m_0 gives $\operatorname{covol}(\mathbb{Z}^3 \cap m_0^\perp) = |m_0|$. Let $g = \gcd(\alpha, \beta, \gamma)$ be the content, $q = g q_0$ with q_0 primitive of discriminant $\Delta_0 = \Delta/g^2$. Then $r_{L_m}(D) = r_q(D) = r_{q_0}(D/g)$ (zero unless $g \mid D$), and $d(D/g) \leq d(D)$, so it suffices to bound r_{q_0} .

2-adic control (proved). A primitive m_0 has at least one odd coordinate, and a case check mod 8 (one, two, or three odd coordinates give $|m_0|^2 \equiv \text{odd}, 2 \text{ or } 6 \pmod{8}, \text{ odd}$, respectively) shows $\nu_2(|m_0|^2) \leq 1$. Hence $\nu_2(\Delta_0) \leq \nu_2(\Delta) \leq 3$: the prime 2 is uniformly bounded-ramified and contributes only an absolute constant below. The ramified/unramified dichotomy is therefore a statement about ODD primes only.

3. Lemma NT-1: representations correspond to square roots of Δ_0

Lemma NT-1. *Let q_0 be a primitive positive-definite integral binary form of discriminant $\Delta_0 < 0$ and $n \geq 1$. Let $r_{q_0}^*(n)$ count the representations $q_0(a, b) = n$ with $\gcd(a, b) = 1$. Then*

$$r_{q_0}^*(n) \leq w(\Delta_0) \cdot N_{\Delta_0}(n), \quad N_{\Delta_0}(n) = \#\{s \bmod 2n : s^2 \equiv \Delta_0 \pmod{4n}\},$$

where $w(\Delta_0)$ is the number of proper automorphs: 6 if $\Delta_0 = -3$, 4 if $\Delta_0 = -4$, 2 otherwise.

Proof. Let $q_0(a, b) = n$ with $\gcd(a, b) = 1$. Choose $c, d \in \mathbb{Z}$ with $ad - bc = 1$ and set $U = \begin{pmatrix} a & c \\ b & d \end{pmatrix} \in SL_2(\mathbb{Z})$. Then $(q_0 \circ U)(x, y) = nx^2 + sxy + ty^2$ for some $s, t \in \mathbb{Z}$ with $s^2 - 4nt = \Delta_0$, so $s^2 \equiv \Delta_0 \pmod{4n}$. The completion (c, d) is unique up to $(c, d) \mapsto (c + \lambda a, d + \lambda b)$, which shifts $s \mapsto s + 2\lambda n$: the class $s \bmod 2n$ depends only on the representation. This defines $\varphi : \{\text{primitive representations}\} \rightarrow \{s \bmod 2n : s^2 \equiv \Delta_0 \pmod{4n}\}$. If $\varphi(a, b) = \varphi(a', b')$, the corresponding U, U' satisfy $q_0 \circ U = q_0 \circ U'$ (the middle coefficient fixes $t = (s^2 - \Delta_0)/4n$), so $U'U^{-1}$ is a proper automorph of q_0 carrying $(a, b)^\top$ (first column of U) to $(a', b')^\top$. A positive-definite form has exactly $w(\Delta_0)$ proper automorphs (the classical count via the reduced representative: $\pm I$ always; the extra ones only for $\Delta_0 = -3, -4$). Hence every fiber of φ has at most $w(\Delta_0)$ elements. $\square$

4. Lemma NT-2: counting the square roots, and the unramified assembly

Lemma NT-2. *For $M \geq 1$ and any integer Δ ,*

$$\#\{x \bmod M : x^2 \equiv \Delta \pmod{M}\} \leq 2^{\omega(M)+1} \prod_{p \text{ odd}, p \mid \gcd(M, \Delta)} p^{\lfloor \min(\nu_p(\Delta), \nu_p(M))/2 \rfloor},$$

where ω counts distinct prime factors.

Proof. By CRT it suffices to bound the roots mod p^k , $p^k \parallel M$. Let $\nu = \min(\nu_p(\Delta), k)$ (with $\nu_p(0) = \infty$). p odd: if $\nu = 0$, Hensel lifting of the at most 2 roots mod p gives ≤ 2 . If $\nu = k$, $x^2 \equiv 0 \pmod{p^k}$ forces $\nu_p(x) \geq \lceil k/2 \rceil$: $p^{\lfloor k/2 \rfloor}$ roots. If $0 < \nu < k$: solutions require ν even and $x = p^{\nu/2}u$ with $u^2 \equiv \Delta/p^\nu \pmod{p^{k-\nu}}$, a unit square: ≤ 2 Hensel roots, each giving $p^{\nu/2}$ choices of $x \bmod p^k$ — total $\leq 2p^{\nu/2}$. In all cases the count is $\leq 2p^{\lfloor \nu/2 \rfloor}$. For $p = 2$ the same analysis with the usual mod-8 care gives $\leq 4 \cdot 2^{\lfloor \nu/2 \rfloor}$, absorbed into the leading $2^{\omega(M)+1}$ together with one factor 2 per odd prime. $\square$

Unramified corollary (explicit constant). Suppose every odd $p \mid n$ has $p^2 \nmid \Delta_0$. For odd p : $\nu_p(\Delta_0) \leq 1$, and an odd $\nu < k$ admits no solution, so each odd prime contributes ≤ 2 . For $p = 2$: $\nu_2(\Delta_0) \leq 3$ (Sec. 2), contributing $\leq 4 \cdot 2^{\lfloor 3/2 \rfloor} = 8$. Since $N_{\Delta_0}(n) = \frac{1}{2} \#\{x \bmod 4n : x^2 \equiv \Delta_0 \pmod{4n}\}$ (the two lifts $s, s + 2n$ of each class coincide mod $4n$ -condition), we get $N_{\Delta_0}(n) \leq \frac{1}{2} \cdot 8 \cdot 2^{\omega_{\text{odd}}(n)} \leq 4d(n)$, hence by NT-1

$$r_{q_0}^*(n) \leq 6 \cdot 4d(n) = 24d(n).$$

Removing primitivity: every y with $|y|^2 = D$ is $y = e y_0$ with y_0 primitive and $e^2 \mid D$, so

$$r_{q_0}(D) = \sum_{e^2 \mid D} r_{q_0}^*(D/e^2) \leq 24 \sum_{e^2 \mid D} d(D/e^2) \leq 24d(D) \cdot \#\{e : e^2 \mid D\} \leq 24d(D)^2.$$

With Sec. 2 (content) this proves Lemma NT(i): $r_{L_m}(D) \leq 24d(D)^2 \ll_\varepsilon D^\varepsilon$. $\square$

5. The ramified corner: citation, outline, exhaustive applied coverage

Statement (imported). For odd p with $p^2 \mid \Delta_0$ and $p \mid D$ the same conclusion holds; classically, for every positive-definite integral binary form one has the sharp uniform bound $r_{q_0}(D) \leq 6d(D)$ (Dirichlet's

representation formula: the total count over all classes of discriminant Δ_0 is $w(\Delta_0) \sum_{d|D} \chi_{\Delta_0}(d)$ for the fundamental part, extended to non-fundamental discriminants by conductor reduction; a single class is at most the total). Book-level sources: Landau, *Vorlesungen über Zahlentheorie* (Part IV, representation numbers of binary forms); Cassels, *Rational Quadratic Forms* (representation theory of binary forms and conductor descent); Cox, *Primes of the Form $x^2 + ny^2$* (Dirichlet formula, fundamental case).

Descent outline (not load-bearing). Over $\mathbb{Z}_p$ (p odd, $p^2 \mid \Delta_0$) diagonalize $q_0 \cong \varepsilon_1 x^2 + \varepsilon_2 p^j y^2$, $j = \nu_p(\Delta_0) \geq 2$, ε_i units. A representation of D with $p \mid D$ and $p \nmid x$ would force $\varepsilon_1 x^2 \equiv D \pmod{p}$, fixing $x \pmod{p}$ up to sign; iterating localizes the count, and extracting $x = px'$ where forced reduces q_0 to a form of discriminant Δ_0/p^2 (conductor descent), terminating in the unramified case of Sec. 4 after $\leq \nu_p(\Delta_0)/2$ steps with a bounded multiplicative loss per step.

Exhaustive applied-domain machine coverage (load-bearing for the package). The new script `codes/vacuum/dr2_lemma_nt_exhaustive.py` (5/5 asserts, 1.2s) verifies the applied chain $r_Q(m) \leq 6d(4R - |m|^2)$ for EVERY distinct sum $m \neq 0$ of every shell used by the DR-2 package: 2,644,976 sums across $R = 101/314/909/1826/4994/9974$, global worst ratio 0.250; degenerate $D = 0$ sums have $r_Q(m) = 1$ exactly; the 24,096 ramified-corner instances (odd p : $p^2 \mid |m_0|^2$, $p \mid D$) ALL satisfy the bound with worst ratio 0.119; and the per-shell maxima reproduce the archived T' values (18/24/24/24/32/48). Thus no instance to which the DR-2 theorem is actually applied rests on the citation alone.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “The in-bundle constant is $24d(D)^2$, but the package quotes $6d(D)$.” VALID with mitigation: the DR-2 conclusion needs only $T' \ll_\varepsilon R^\varepsilon$, for which $d(D)^2 \ll_\varepsilon D^\varepsilon$ is fully sufficient; the package v1.4 presents the in-bundle-proved form as load-bearing and the sharp $6d(D)$ as the classical remark. No claim depends on the constant 6.
- (β) “The ramified corner is still an import.” VALID with mitigation, by design (operator two-tier directive): the corner is (a) stated precisely, (b) outlined, (c) book-level cited, and (d) exhaustively machine-verified on the ENTIRE applied domain (24,096 instances, worst 0.119) — the only place the citation is load-bearing is the abstract uniform statement beyond the applied shells.
- (γ) “The $\nu_2(|m_0|^2) \leq 1$ and covolume $= |m_0|$ steps are asserted.” DISMISSED: both are proved in Sec. 2 (mod-8 case check; $\mathbb{Z}^3 = L_{m_0} \oplus \mathbb{Z}w$ with $w \cdot m_0 = 1$ for primitive m_0 gives $\text{covol} = |m_0|$), and both are implicitly machine-exercised by the exhaustive script across all 2.6M sections.

Quantitative sanity check (mandatory). The NT-1 fiber bound is checked against the data: worst observed $r_Q(m)/6d = 0.250$, i.e. a $\times 4$ headroom; the unramified constant evaluates to $24d(D)^2 \geq 6d(D)$ always (the proved bound is weaker than the cited sharp one, never stronger — the safe direction); and the exhaustive script’s per-shell maxima cross-match the independent `dr2_lattice_divisor.py` archived T' values exactly.

7. Result footer

Result ID: B5-BEYOND-LAYER-BOUND / DR-2 Lemma NT
in-bundle pin

Precise statement: (i) UNRAMIFIED (proved in full, elementary):
 $r_{\{L_m\}}(D) \leq 24 d(D)^2 \ll_{\varepsilon} D^{\varepsilon}$ via
NT-1 (primitive reps $\leftrightarrow$ square roots of
 $\Delta_0 \pmod{4n}$, fiber $\leq w(\Delta_0) \leq 6$)
+ NT-2 (CRT/Hensel root count) + content/
imprimitivity reduction + $\nu_2(|m_0|^2) \leq 1$.
(ii) RAMIFIED corner (odd $p^2 \mid \Delta_0$, $p \mid D$):

	classical 6 $d(D)$ cited (Landau/Cassels/Cox) + descent outline + EXHAUSTIVE applied- domain machine check (2,644,976 sums, all shells, worst 0.250; 24,096 ramified instances, worst 0.119).
Scope:	rank-2 sections L_m of Z^3 (DR-2 lattice class). The DR-2 chain needs only $d(D)^2 \ll_{\text{eps}} D^{\text{eps}}$: tier (i) suffices for every unramified instance; tier (ii) is machine-covered on the entire applied domain.
Evidence grade:	ANALYTIC (NT-1, NT-2, unramified assembly: complete proofs) + EXECUTED (dr2_lemma_nt_exhaustive.py 5/5, 1.2 s) + CITED (ramified corner, book-level).
Reproduction command:	python codes/vacuum/dr2_lemma_nt_exhaustive.py ; codes/vacuum/dr2_divcirc_proof.py
Falsifier:	any m, D with $r_{\{L_m\}}(D) > 24 d(D)^2$ in the unramified case; any applied-domain sum with $r_Q(m) > 6 d(4R - m ^2)$; a counter- example to the NT-1 fiber bound.
Tier before / after:	supports the DR-2 final T7 decision: the Lemma NT pin requested by the operator verdict 2026-06-12 (in-bundle proof route). No standalone tier; arbiter is the DR-2 package + claim ledger.
No-overclaim statement:	tier (i) is unconditional and self- contained; tier (ii) remains an import beyond the applied domain; the sharp constant 6 is NOT claimed in-bundle.
Next required action:	operator review of this note -> decide the DR-2 final T7 promotion (and whether the ramified corner's citation suffices or the conductor descent should be written out).